If the AI Bubble bursts, will You Win or Lose?

Why some organisations will thrive after the AI-Gold Rush, and others won't

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Introduction

In the summer of 1845, the *Railway Gazette* ran advertisements for 1,200 miles of new British rail lines in a single week. Parliament was approving track faster than engineers could survey the land. Even prominent Victorian families and intellectuals participated in railway speculation. Everybody knew railways would change the world, and they were right. What they got catastrophically wrong was the assumption that being right about technology meant being right about the investment.

Railway share prices collapsed dramatically after 1845, with many losing the majority of their value (Railway Mania). The infrastructure, however, remained. It powered the British economy for a century.

I have been thinking about that distinction between the technology and the financial narrative built around AI. Because right now, both things are true simultaneously: we are living through one of the most significant technological transitions in economic history, and we are also, in specific and identifiable segments of the market, in what has characteristics of a speculative bubble. The strategic error most business leaders are making is treating those two statements as contradictory. They are not. Understanding how they coexist, and where the boundary between them runs, is the only analytical question that will determine which companies win the AI decade.

The scale is real. That is precisely the problem.

Start with the level of investment, because they are extraordinary enough to justify the excitement and alarming enough to justify caution simultaneously.

Global investment in artificial intelligence is accelerating at a pace rarely seen in technological history. While exact figures vary by methodology, most estimates place near-term AI-related spending in the hundreds of billions of dollars annually, with rapid growth expected throughout the decade.

AI has also become one of the most concentrated areas of venture capital investment. According to the Stanford AI Index, AI-related startups now account for a significant and rapidly growing share of global VC funding, particularly in the United States.

Meanwhile, major technology companies, including Microsoft, Amazon, Alphabet, and Meta are collectively investing tens of billions of dollars each year in AI infrastructure, data centers, and compute capacity. These investments are scaling rapidly as demand for large-scale models and AI services expands.

Broader public and private initiatives, including semiconductor and compute infrastructure programs such as the US CHIPS and Science Act, are reinforcing this trend by accelerating capital flows into the AI supply chain.

Taken together, these investments represent one of the largest coordinated capital shifts into a single technological domain in modern economic history.

At the frontier of AI, valuations are already pricing in extraordinary expectations: private AI leaders are widely estimated to generate billions in annual revenue, yet command valuations that imply extreme revenue multiples compared to traditional software norms. Even public market leader Nvidia has, at times, traded at elevated levels that would once have been considered unsustainably rich. The signal is clear: capital is no longer pricing AI as a product category, but as foundational economic infrastructure.

AI infrastructure investment has become one of the most important drivers of recent US economic growth, with data center and technology capex making a visibly outsized contribution to GDP expansion in recent quarters. Economists like Jason Furman and others have highlighted how unusually concentrated this growth has become in a narrow set of AI-linked investments. In effect, parts of the US economy are increasingly tethered to the continuation of the AI investment cycle.

Senior leaders today often fall into two equally risky instincts when confronted with the scale of investment in AI.The first is dismissal. Because the levels of capital flowing into the space can appear irrational or inflated, the conclusion becomes to wait on the sidelines until the picture “clears.” This stance risks underestimating structural change by treating momentum as noise.

The second is capitulation. Here, the sheer magnitude of investment is interpreted as inevitability—proof that action must be immediate and aggressive to avoid being left behind. This leads to rushed commitments that are not always grounded in clear strategic intent.

Both responses are understandable but ultimately flawed. A more productive analytical stance requires stepping back and disaggregating what we actually mean by “AI investment.” It is not a single signal. It spans infrastructure, model development, application layers, integration efforts, and speculative capital flows that behave very differently from one another.

Without that distinction, leaders react to a headline number rather than a structured reality. What’s needed instead is a clearer map of where value is being created, where it is being anticipated, and where it is merely being priced in.

What History actually teaches (It is not what you think)

The Dotcom bubble remains the clearest historical reference point for today’s AI-driven market cycle, though often misunderstood in hindsight. Between the mid-1990s and the 2000 peak, the Nasdaq surged roughly 400–500%, while most tech IPOs were unprofitable and valuations detached from fundamentals. When the correction arrived, an estimated $5 trillion in market value was erased, with iconic names like Pets.com and Boo.com collapsing almost as quickly as they had risen.

But here is what the Dotcom narrative gets wrong when applied to AI: the bust did not slow the internet. It accelerated it. The fibre-optic cables laid at enormous private loss in the late 1990s became the cheap bandwidth infrastructure that enabled Google, Amazon, and the entire 2010s platform economy. The

investors lost. Technology won. The companies that built durable businesses were not the ones that raised the most money in 1999, they were the ones that used the cheap post-bust infrastructure to serve genuine demand that the speculation had correctly, if prematurely, identified.

The electrification parallel is more instructive still. When electrical infrastructure was built in the 1880s and 1890s, productivity statistics barely moved. Factories were wired, but they were still organised around the logic of steam, a central shaft driving mechanical belts to individual machines. It was only in the 1920s, when factories were redesigned from first principles around distributed electric motors, that the productivity gains showed up in macroeconomic data. The technology worked the whole time. The organisational transformation required to capture its value took a generation.

Erik Brynjolfsson of Stanford has long argued that general-purpose technologies like AI follow a "productivity J-curve": initial disruption and investment often precede measurable economic gains. Recent US productivity data has shown signs of strengthening growth after a prolonged period of stagnation, reinforcing the idea that AI-driven tools may finally be filtering into real economic output. If this pattern holds, we may be entering the phase where the benefits of AI begin to show up in aggregate productivity statistics.

The railway parallel adds a third layer. The Victorian railway network was real, useful, and economically foundational. The financial claims made about it during the mania were not. And critically: the assets that proved durable were not the railway company equities that investors held. They were the physical infrastructure itself, the track, the stations, the engineering knowledge, which served the economy regardless of which companies owned it.

That is the question every business leader should be asking about AI: am I investing in the physical and organisational equivalent of 'railway infrastructure', or in the equity of 'railway company promoters'?

The value chain map: Where value lives and where it is priced on faith

The AI value chain runs 10 layers deep, from raw materials and semiconductor manufacturing through to enterprise applications. Value is not evenly distributed, and history suggests it rarely is during platform shifts.

At the infrastructure base, advantages are genuinely structural. ASML effectively controls extreme ultraviolet lithography, the machines required to produce cutting-edge chips. Every frontier AI chip depends on its technology. TSMC dominates advanced chip fabrication, while NVIDIA captures the bulk of AI accelerator demand in data centres, reinforced by its CUDA software ecosystem, which has become the industry standard.

The hyperscaler layer (Amazon Web Services, Microsoft Azure, and Google Cloud) occupies a historically resilient "picks and shovels" position. They provide compute to all participants, benefit from high switching costs, and can sustain heavy capital expenditure from diversified cash flows.

The foundation model layer is more complex. Companies such as OpenAI and Anthropic are scaling revenue at unprecedented speed for enterprise software. Anthropic, for example, has moved from roughly $100m in 2023 to multi-billion dollar run rates by 2025. However, profitability remains distant, and costs scale with usage due to continuous training and inference demands. Open-source models, including Meta's Llama family, also exert persistent downward pressure on pricing power.

The application layer is where valuation risk is most concentrated. Many “AI companies” are effectively wrappers around third-party models, without proprietary data, workflow integration, or switching costs. These businesses are highly exposed to commoditisation as model capabilities diffuse.

By contrast, durable application-layer value is likely to sit in sectors where AI is combined with proprietary data and deep domain expertise — healthcare, legal, industrial systems, and specialist financial workflows. In these areas, AI is not the product; it is the enabling layer.

The Economists are not agreeing on AI’s economic impact. That is the point.

Four of the most serious economists working on technology and markets have staked distinct positions on AI's economic impact, and their disagreement is not noise, it is signal about the sincere uncertainty business leaders need to price into their decisions.

MIT economist and 2024 Nobel laureate Daron Acemoglu offers a notably restrained macro view of AI. His modelling suggests a US GDP uplift of roughly 1.1%–1.6% over the next decade, equating to very modest annual productivity gains of around 0.05%. His concern is less about adoption and more about design: “so-so automation” that cuts labour costs without materially improving output, self-checkouts or basic customer service automation being typical examples.

Robert Shiller, who called both the Dotcom bubble and the 2008 housing crisis, looks at the current Shiller CAPE ratio and notes that the only time it has been higher was during the Dotcom peak of 1999.

Paul Krugman calls the AI boom's market behaviour "unmistakably similar" to the late 1990s tech mania and adds a constraint that financial analysts tend to ignore: the physical infrastructure limits of AI at scale. Data centres require water for cooling at volumes that cannot simply be scaled by capital investment. Energy demand is reshaping national grid planning. These are not soft concerns; they are hard ceilings that will constrain the AI buildout regardless of what financial markets believe.

Erik Brynjolfsson disagrees with the pessimists on timeline, and his 2025 productivity data gives him grounds to. But, and this is the point that gets lost in the debate, Brynjolfsson's optimism about the technology is not optimism about current valuations. Productivity gains accruing to the broader economy do not automatically translate into returns for investors who bought foundation model equity at 60 times revenue.

The practical synthesis of these four frameworks, for a business leader, is this: the technology will deliver real economic value, probably on a longer timeline than markets are currently pricing, and the organisations that will capture that value are not primarily the ones making the largest financial bets, they are the ones making the most intelligent organisational transformations.

The China variable changes the global equation

Everything above assumes a US-centric frame of reference. That frame is insufficient.

DeepSeek’s R1 reasoning model highlighted how rapidly efficiency innovation is reshaping the AI landscape, showing that strong performance can emerge with far lower training cost and compute than previously assumed. While benchmark results are task-dependent, its release intensified debate around

whether raw compute alone remains the dominant driver of frontier capability. Market reactions reflected this uncertainty, underscoring how sensitive AI valuations have become to shifts in perceived technological efficiency.

China's AI development model differs from the US model in three fundamental ways that directly affect how business leaders outside both countries should position themselves. First, it is state-led and strategically patient, meaning it is structurally insulated from the financial bubble dynamics that threaten the US cycle. Second, it is open-source and deployment-oriented, meaning capability is being diffused into the global economy at near-zero cost rather than monetised at premium prices. Third, it is explicitly oriented toward application and industrial integration, the layer the US model has most consistently underinvested in relative to frontier model development.

Chatham House makes the strategic implication explicit: a US financial bubble correction would not slow global AI development, it would shift the centre of gravity toward China. The real competition, as Harvard's Graham Allison framework suggests, is not about which country builds the most capable model. It is about whose AI infrastructure achieves the broadest global adoption. DeepSeek V4, released in April 2026 and optimised for Huawei's Ascend chips rather than Nvidia's, is a data point in that adoption contest that has nothing to do with US financial markets and everything to do with which technology stack the Global South, Southeast Asia, and emerging market enterprises build on over the next decade.

The Observer Research Foundation has identified a dynamic that should concern every business leader thinking about AI capital allocation: the US-China rivalry is itself inflating the US AI bubble. Investment decisions that would be questioned purely on financial grounds are being justified by strategic necessity "we cannot fall behind China." This geopolitical premium on spending makes the bubble harder to deflate and the eventual correction potentially sharper, because rational financial caution is being systematically overridden by strategic anxiety.

What you should actually do

None of this analysis paralyses action. It sharpens it.

Build the capability, not the position. Every scenario in serious analysis-controlled correction, sustained transformation, partial bust, systemic shock, rewards organisations that have embedded genuine AI capability into their operations and penalises those that have primarily acquired financial exposure to AI companies. The technology will outlast any financial correction. Your organisation's AI literacy, proprietary data assets, and redesigned workflows will compound in value regardless of what happens to foundation model valuations. Invest there first.

Treat your proprietary data as a strategic asset, not an operational byproduct. Acemoglu's "so-so automation" failure mode is almost always a data problem: AI deployed on generic models against generic data produces generic results. The organisations achieving measurable productivity gains from AI, and they exist in growing numbers, are almost invariably doing so by combining external model capability with proprietary data and domain expertise that no one else has. Building, curating, and governing that data is not an IT project. It is the central strategic investment of the AI decade.

Diversify your AI vendor relationships deliberately. The report's value chain analysis identifies NVIDIA's CUDA ecosystem as a genuine switching-cost moat, but DeepSeek's architectural innovations and Huawei's Ascend platform are actively contesting that position at state expense. Any organisation that has built its AI infrastructure around a single hardware or model provider is carrying concentration risk

that most board risk registers have not yet captured. Build multi-vendor architecture now, while it is a strategic choice rather than an emergency response.

Apply different risk frameworks to different layers of the value chain. Exposure to semiconductor infrastructure through suppliers, through procurement relationships, through understanding chokepoint dynamics, carries different risk to exposure to foundation model labs, which carries different risk again to application-layer point solutions. Most organisations are applying a single "AI investment" mental model across all of these. That conflation is where capital gets misallocated.

Watch China's deployment, not its benchmarks. The relevant question about Chinese AI for most business leaders is not whether DeepSeek V4 outscores GPT-5 on a reasoning benchmark. It is whether cheap, capable, open-source Chinese AI models achieve adoption in your supply chain partners, your market competitors in Asia, or your industry's emerging-market customer base. That adoption dynamic will reshape competitive landscapes in ways that have nothing to do with the US financial market cycle and everything to do with the cost structure of intelligence in global commerce.

The railway investors of the 1840s were not wrong about railways. They were wrong about which part of the railway story they were actually invested in. The technology rewarded the economy. The mania rewarded the promoters. The serious long-term value accrued to the manufacturers, the merchants, and the consumers, the people who used the infrastructure rather than owned the equity.

The companies that will define the next decade are not the ones that bet most aggressively on AI. They are the ones that understood, clearly and early, that building on the infrastructure of a technological revolution and owning a financial stake in the narrative around it are two entirely different propositions and acted accordingly.

See below for a self-assessment of your current AI strategic position and discover what you need to do to generate value with AI and to survive a potential bubble.

About the Author

Koen Klokgieters is the founder of InnoPlot and has spent more than three decades working at the intersection of strategy, innovation, technology and organisational transformation. His experience combines corporate leadership, consulting, entrepreneurship and education.

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